

A Coverage-Aware High-Quality Sensing Data Collection Method in Mobile Crowd Sensing

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Abstract—In this paper, we leverage unmanned aerial vehicles (UAVs) to enhance mobile crowd sensing (MCS) by addressing two critical challenges: uncontrollable data quality and inevitable unsensed points of interest (PoIs). We introduce a UAV-assisted method to deal with these challenges. To ensure the accuracy of sensing data contributed by human participants, the proposed truth discovery method utilizes UAV-collected sensing data as few-shot samples to train the truth discovery model, which is then employed to calibrate sensing data solely collected by human participants. Additionally, to meet the sensing coverage requirement, we present a method that predicts data values for unsensed PoIs by utilizing their historical sensing data and the sensed neighboring PoIs information. The method employs a graph neural network to capture spatio-temporal relationships of the sensing data, facilitating accurate estimation of unsensed PoIs. Through extensive simulations, our approaches demonstrate superior performance compared to existing methods, showcasing the potential of UAV-assisted MCS for overcoming challenges and enhancing data collection efficiency in various domains.

Index Terms—Coverage-aware, mobile crowd sensing, truth discovery, UAV-assisted.

I. INTRODUCTION

WITH the advancement of 5G/6G networks, everyday smart products such as smartphones, notepads, and wearable devices can seamlessly connect to the Internet of Things (IoT) [1]. As a type of multimodal network, the IoT possesses the potential to significantly revolutionize our everyday lives even further. Mobile crowd sensing (MCS), as a pivotal IoT technology, has emerged as a promising paradigm

for data collection [2], [3]. By leveraging the widespread use of mobile devices among individuals, MCS facilitates the gathering and analysis of large-scale data from the physical world [4]. MCS campaigns play a crucial role in supporting development and research across various domains, including smart cities and polymorphic intelligent networks [5], [6].

While MCS offers the potential to gather a substantial amount of sensing data, there are still critical challenges to face. The first significant challenge is uncontrollable data quality [7], [8], [9]. Variations in sensor quality, participant expertise, and environmental noise lead to considerable disparities in the information yielded by collected sensing data. Consequently, the values provided by participants may lack accuracy and at times, even present contradictions.

Several traditional solutions, such as averaging collected sensing data or employing a voting mechanism, have been proposed to address this challenge. However, the averaging methods often fail to accurately represent uneven data distributions and are vulnerable to skewing by outliers. Similarly, the voting methods face issues such as inconsistent data quality, participant diversity, conflict resolution, and the difficulty of selecting appropriate strategies, all of which can lead to inaccuracies and flawed decision-making. Therefore, to ensure the collection of high-quality sensing data, the platform has to incorporate methods of obtaining high-quality data into the MCS campaigns.

The second challenge is inevitable unsensed points of interest (PoIs). During the execution of MCS tasks, there is always a requirement for perceptual coverage. However, the distribution of data collection devices has a significant impact on data collection result, which means the coverage requirement will not be met if some specific area is not covered by participants. Furthermore, some MCS campaigns also need to collect sensing data frequently in a region for a long time, such as noise monitoring [10] or air condition report [11]. It is possible that certain parts of the region were detected in earlier time segments but are not sensed at the moment, which would likewise fail the requirement for sensing coverage.

In recent years, the integration of unmanned aerial vehicles (UAVs) has significantly propelled the advancement of MCS technology [12], [13], [14]. Thanks to their ability to maneuver without hindrance from terrain and structures, as well as their immunity to traffic congestion, UAVs have demonstrated remarkable efficiency in expediting sensing data collection tasks. Moreover, before data acquisition, UAVs can undertake sensor maintenance to ensure the quality and precision of the sensors,

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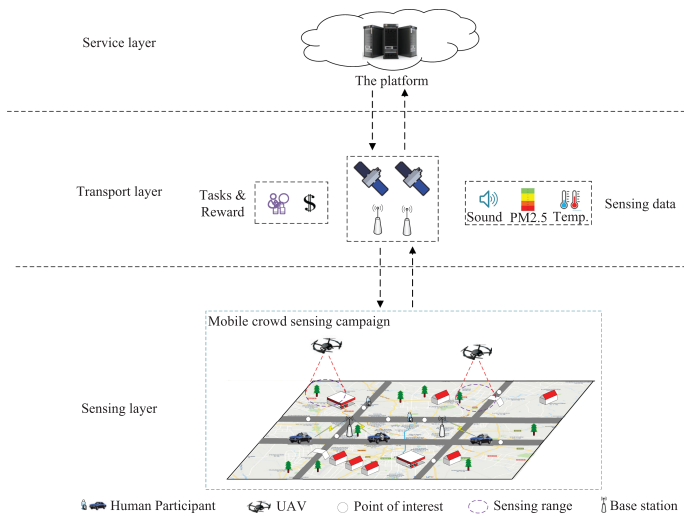


Fig. 1. The proposed hybrid MCS scenario involves the simultaneous collection of sensing data in a region by both UAVs and human participants.

resulting in data of superior accuracy and precision. However, despite these advantages, the reliance solely on UAVs for sensing tasks may not always be feasible due to restrictions or prohibitions on UAV flights in certain urban areas. Consequently, the involvement of human participants remains necessary to complement UAV-based data collection efforts.

The hybrid MCS scenario is illustrated in Fig. 1. In this scenario, MCS tasks require the collection of sensing information in a specific region, such as noise levels and air quality. The platform in the service layer broadcasts these tasks to all human participants in the sensing layer and recruits some of them with rewards through the transport layer. Meanwhile, UAVs, also in the sensing layer and acting as auxiliary components, begin their cruising phase to collect sensing data. The platform coordinates the UAVs' trajectories and transmits movement commands via the transport layer. Once the selected human participants and UAVs have collected the data, they upload it to the platform.

In our previous research [13], we utilized UAVs to calibrate sensing data provided by human participants, employing a process where sensing data collected from the same PoIs by both a UAV and human participant underwent system-level calibration. The calibration frequency aims to enhance the data quality contributed by human participants, as the quality of sensing data relies on the number of calibrations performed. However, despite its effectiveness, this method presents several drawbacks. First, it relies on multiple rounds of data calibration. If a properly calibrated human participant withdraws from an MCS campaign, the entire data collection process will be disrupted, resulting in resource wastage. Moreover, constraints such as battery life and flight restrictions in certain areas pose challenges for UAVs in providing a significant amount of verification data. Finally, the previous research is specifically designed for calibrated types of sensing data. Introducing a new type of data necessitates the execution of a calibration procedure anew.

In the sensing data collection process, besides prioritizing sensing data quality, ensuring sensing coverage is also crucial. In our previous research [14], we utilized UAVs to augment data

collection in areas inaccessible to human participants, thereby enhancing overall sensing coverage. Despite the combined efforts of human participants and UAVs covering most PoIs, there is still a possibility of certain PoIs being unsensed. For instance, certain areas might be temporarily or permanently off-limits due to policies or safety concerns. In such scenarios, conventional sensing data collection methods encounter obstacles, underscoring the significance of devising technical solutions to circumvent these limitations.

In this paper, we tackle the challenges of ensuring high-quality of sensing data and sensing coverage. For the problem of sensing data quality, unlike the traditional truth discovery methods, which need continuous human participant involvement in the sensing data collection process and ensure consistent trust levels throughout. We utilize few-shot samples provided by UAVs to train a truth discovery model, which is then employed to calibrate the sensing data of PoIs sensed solely by human participants, even when their trustworthiness is uncertain. Additionally, to address scenarios with limited UAV sensed data availability, we equip our model with the capability to rapidly adapt to tasks. Furthermore, the model is trained to swiftly adapt to new tasks and yield accurate predictions even with minimal training data.

To address the issue of incomplete sensing coverage, the proposed method leverages historical sensing data and data from nearby sensed PoIs to predict unsensed PoIs data. We quantify and represent the correlations between these PoIs through an adjacency matrix. Given the superiority of graph neural network (GNN) in processing graph-structured data, it is capable of effectively capturing and utilizing the complex relationships between nodes. Therefore, we use an improved GNN to capture and utilize these relationships, effectively predicting the unsensed PoIs data. The core idea is to organize the sensing data of PoIs and sensing time segments into a spatio-temporal information matrix, then serve the matrix as input for the GNN to capture spatio-temporal relationships. Regarding spatial relevance, the method regards PoIs as nodes within a graph, with the relationships between these PoIs serving as edges of the graph. The edge weights are simulated based on geographical location-related information.

The primary innovations of this paper include:

- We present a comprehensive framework that incorporates UAVs into MCS. By leveraging UAV technology, the framework addresses significant challenges related to data quality and coverage in MCS applications, providing a systematic approach to enhance data collection efficiency and accuracy.
- We introduce an innovative truth discovery method based on few-shot samples (F-TDM) that leverages high-quality data collected by UAVs to refine and enhance sensing data contributions. Unlike previous calibration methods, which required UAVs to frequently interact with human participants to adjust sensing parameters, F-TDM uses a few-shot learning-based truth discovery model. This model calibrates data by training on a limited set of UAV-collected samples, enabling it to adapt easily to new data types and environments without direct UAV-human participant interaction. This approach offers greater flexibility and

reduces dependency on UAV encounters, thus enhancing adaptability and efficiency.

- To meet sensing coverage requirement, the paper proposes unsensed PoI data prediction method named UPM that estimates values for unsensed PoIs by analyzing historical data and spatial relationships among PoIs. Utilizing a graph neural network, the UPM captures intricate spatio-temporal correlations, enabling accurate predictions even in areas with sparse data. The method significantly improves data completeness and supports effective decision-making in various MCS applications.
- We present extensive simulation results based on the real data to verify that the proposed method performs more accuracy compared to other state-of-the-art methods.

The remaining part of this paper is organized as follows. We introduce related work in Section II. We describe the system model in Section III. The few-shot samples based truth discovery method and unsensed PoI prediction method are proposed in Sections IV and V, respectively. Last, the experimental analysis is given in the Section VI. Finally, the conclusion is presented in the Section VII.

II. RELATED WORK

In this section, we briefly review the related work on the truth discovery and data quality, task allocation for MCS, and unsensed PoIs data prediction, respectively.

A. Truth Discovery and Data Quality

In MCS, truth discovery has attracted widespread attention as an important theory and various approaches have been proposed, which are designed to tackle the challenge of extracting true information from unreliable sensing data. Articles [15], [16], [17], [18] utilized classical truth discovery protocols following a sequential process of iteratively estimating user weights (weight estimation) and task veracity (truth estimation) until a predefined convergence criterion was met (convergence determination). They calculated weights based on the distance between the data observed by human participants and the ground truth. Then, in the truth estimation phase, the weights of human participants were fixed to estimate the veracity of the perception task.

Sun et al. [19] proposed a truth discovery method that assumed the errors between the data submitted by human participants and the ground truth followed a Gaussian distribution. Based on the expected overall quality of the human participants, the method iteratively performed truth inference and weight estimation. If a human participant's submitted sensing data aligned more closely with the estimated ground truth, they would be assigned a higher weight until a convergence criterion was met. Ye et al. [20] formulated the role of relationships between entities in truth discovery tasks as a constrained optimization problem. They proposed algorithms to partition the entities into disjoint groups and generate arithmetic constraints for each disjoint group separately. Li et al. proposed a privacy-preserving truth discovery mechanism for continuous data aggregation, which involved each human participant independently sampling private noise from a private noise distribution and adding it to their data. The

server then performed weighted aggregation of the information from each human participant to obtain the final output. Mo et al. [21] used historical data to verify reported data, gain human participant trust, and selects the most reliable ones for data perception tasks.

Data quality refers to the accuracy, completeness, and reliability of the data collected from human participants' mobile devices or sensors. It is of utmost importance in MCS as it directly influences subsequent data analysis and decision-making outcomes. Zhao et al. [22] proposed an incentive-based mechanism that compelled self-interested human participants to truthfully report objectively accurate perception observation. They also introduced proactive perception bias calibration by human participants to enhance the quality of sensing data from the source. Zhao et al. [23] proposed a zero-knowledge model for data reliability assessment, quantifying the data quality based on the deviation between reliable data and ground truth values. The incentive rewards were provided to human participants based on data quality. Yang et al. [24] proposed a decentralized trust inference approach that utilized UAV sensing data as a cornerstone for trust evaluation in zero-trust MCS. They established trust baselines at different levels to evaluate the trustworthiness of human participants. Then, by prioritizing the selection of reliable and low-bid human participants, they aimed to enhance data quality and reduce costs. Jiang et al. [25] proposed a human participant selection scheme based on service quality, utilizing the reputation and willingness of human participants, by considering their historical task completion records. They established a service quality model and used it to improve the task quality.

B. Task Allocation

With the growing popularity of UAVs, they are playing an increasingly pivotal role in the field of MCS. Proper task allocation for UAVs is crucial in MCS. Allocating UAVs to different areas enables wider area coverage, higher efficiency in collecting perception data, and ensures balanced data collection across the entire region or community. Considering factors such as UAV flight time, battery life, and sensor capacity, a rational allocation of UAV resources and capabilities is essential. Maximizing resource utilization and task execution efficiency can be achieved by assigning tasks to the most suitable UAV.

Dai et al. [26] addressed delay-sensitive UAV MCS scenarios by proposing a centralized control and distributed execution framework based on distributed deep reinforcement learning. The authors aimed to optimize the data collection rate while minimizing UAV power consumption and ensuring data freshness throughout each time period. Zhou et al. [27] proposed a joint optimization approach for UAV-assisted MCS route planning and task allocation. Their objective was to minimize UAV energy consumption while maximizing the benefits of the MCS carrier. Xu et al. [28] proposed a task allocation method for MCS and introduced three algorithms designed to minimize total sensing cost, average task cost, and total travel distance for UAVs, while maximizing the UAVs' overall contribution. Deng et al. [29] proposed a task allocation method that facilitates the sharing of

task state information among UAVs, introducing a multi-agent deep reinforcement learning algorithm to address UAV task allocation and path planning challenges. Xu et al. [30] studied the UAV task allocation problem and proposed a reverse-auction-based distributed task assignment incentive mechanism, aimed at encouraging UAVs to cooperatively participate in sensing tasks. Song et al. [31] investigated the cooperative path planning problem for vehicles and UAVs. The authors employed genetic algorithms and reinforcement learning to optimize UAV path planning, addressing both route planning and task allocation jointly. Liu et al. [32] examined task allocation issues for UAVs and human participants within UAV-assisted MCS. They formulated the problem as a Markov decision process and proposed a real-time solution based on proximal policy optimization to address these challenges.

In the area of task allocation for human participants, Wang et al. [33] introduced a distributed task allocation scheme for MCS that leverages route navigation systems. The study addresses both time-tolerant and time-sensitive tasks, formulating them as potential games to balance the preferences of human participants with the platform's requirements. Wei et al. [34] examined the task allocation problem in Sparse MCS and proposed a framework that utilizes Gaussian Processes for selecting human participants and allocating sensing tasks. Zhang et al. [35] proposed a multi-task allocation method in MCS that utilizes human participants' historical trajectories to predict their locations, allocating tasks through a global heuristic search algorithm. Tan et al. [36] explored the problem of multiple cooperative task allocations in social MCS, leveraging mobile social networks for human participant recruitment. Their method utilizes real-life relationships within social networks to form compatible groups of human participants.

C. Unsensed PoIs Data Prediction

In MCS, the issue of unsensed data prediction has received widespread attention. Tian et al. [37] took into account the outlier data uploaded by the human participants and GPS location errors, and proposed a grid distance-based tensor completion unsensed data inference algorithm. Initially, the proposed algorithm applied an outlier data processing method to eliminate any outlier values in the data. It then employed a GPS location error processing method to compute more precise environmental data for each time slot and grid. Finally, it used an unsensed data inference method to obtain comprehensive environmental data for the entire sensing region. Wang et al. [38] proposed a generative model aimed at inferring unsensed data, which combined the traditional deep matrix factorization and generative adversarial networks to fill in the sensed data. The model learned the spatio-temporal features of the sensing data and employed binary cross entropy loss predict the unsensed data. Zhang et al. [39] proposed a deep learning method to tackle the issue of unsensed data in the construction of MCS radio environment maps. They employed a variational auto-encoder to infer the unsensed data. Wang et al. [40] proposed a deep learning industrial sensing and prediction scheme based on sparse MCS to predict unsensed data. The process began by using the deep matrix factorization method to restore the current full sensing map, then captured and exploited

the spatio-temporal correlation to predict the unsensed data. Wang et al. [41] proposed a matrix completion algorithm based on a bipartite graph with spatio-temporal constraints to restore complete data. Then, the graph convolutional networks were used for unsensed data prediction. Yang et al. [42] proposed a Spatial-Temporal Graph Attention (STGA) model that leveraged graph attention networks with spatio-temporal constraints to reconstruct unsensed sensor data in sparse mobile crowdsensing. The model uses a message-passing mechanism to learn signal correlations and employs a hierarchical attention mechanism to select and weight high-contribution nodes, improving the accuracy of the reconstructed graphs.

Compared to previous state-of-the-art techniques, we propose a novel truth discovery method that addresses the issue of fixed human participant reliability in traditional truth discovery. By employing meta-learning techniques, we have devised a new approach to obtaining high-quality data in crowd sensing, along with appropriate UAV task allocation incorporated into the method. For unsensed PoIs data prediction issues, we construct a graph neural network model, utilizing historical data and information from nearby PoIs to predict the unsensed PoIs data.

III. SYSTEM MODEL

Consider a UAV-assisted MCS system. Data consumers publish a series of data collection tasks through the MCS system, setting corresponding rewards for each task. These tasks might include air quality monitoring, humidity sensing data collection, and more. Each task may involve the collection of data from certain PoIs within a specified range during particular time periods. Given the unknown quality of data collected by human participants, data consumers require multiple human participants to collect data for a single task multiple times to comprehensively evaluate the required data. UAVs typically have higher accuracy and precision, thus data consumers can involve UAVs in data collection to enhance data quality. However, due to flight restrictions and other limitations, UAVs can only complete a limited number of tasks in certain scenarios. Generally, this means that without UAV participation, the quality of the collected data cannot be significantly improved.

In the UAV-assisted MCS system proposed in this paper, a small number of UAVs can be utilized to quickly train a truth discovery model by participating in data collection tasks. This model can then calibrate the data points collected solely by human participants, thereby improving the data quality. This process requires only a minimal additional cost, as detailed in the Section VI. Beyond improving data quality, the UAV-assisted MCS system can also predict data of PoIs that data consumers need but are not collected by human participants due to inaccessibility or insufficient reward incentives.

A series of tasks are denoted by $\mathcal{T} = \{t | t = 1, 2, \dots, T\}$. Each task t requires to sense several PoIs during N time segments. The set of PoIs are defined by $\mathcal{P} = \{p | p = 1, 2, \dots, P\}$. There are human participants who are ready to engage in MCS campaigns, which are denoted by $\mathcal{I} = \{i | i = 1, 2, \dots, I\}$. Additionally, there are also a series of UAVs acting as

TABLE I
LIST OF IMPORTANT NOTATIONS

Notation	Explanation
$\mathcal{T}$	Set of MCS Tasks
$\mathcal{P}$	Set of PoIs
$\mathcal{I}$	Set of human participants
$\mathcal{U}$	Set of UAVs
N	Number of time segments in matrix
$\mathbf{X}_O$	Original Matrix
$\mathbf{X}_R$	Real Matrix
$\mathbf{M}$	Mask Matrix
$\mathbf{Y}$	Inference Matrix
$\mathbf{D}_W$	Data Matrix provided by human participants
d_t	Ground truth

participants to provide sensing data, which are defined by $\mathcal{U} = \{u|u = 1, 2, \dots, U\}$.

In addressing the challenge of prediction for unsensed PoIs, we consider that there are P PoIs needed to be sensed, and each sensed PoI comprises a series of time-segmented data, which could be organized into a matrix with P rows (representing sensed PoIs) and N columns (representing time segments). Due to the inability to collect certain PoIs during specific time segments, the matrix needs to be inferred to improve its completeness. We define several related matrices in the subsequent discussion.

Definition 1 (Original Matrix): A matrix composed of P sensed PoIs and N time segments, denoted as $\mathbf{X}_O \in \mathbb{R}^{P \times N}$. In this matrix, $\mathbf{X}_O[i, j], i \in [0, P], j \in [0, N]$ represents the real data of the i -th sensed PoI during the j -th time segment.

Definition 2 (Mask Matrix): A matrix of the same dimensions as the original matrix, denoted by $\mathbf{M} \in \mathbb{R}^{P \times N}$. In this matrix, each element $\mathbf{M}[i, j], i \in [0, P], j \in [0, N]$ indicates whether the data of the i -th sensed PoI during the j -th time segment. If the data is unsensed, $\mathbf{M}[i, j] = 0$; otherwise, $\mathbf{M}[i, j] = 1$.

Definition 3 (Real Matrix): The collected matrix is also the input matrix for our model, which is obtained through the Hadamard product of the original matrix and the mask matrix, i.e., $\mathbf{X}_R = \mathbf{X}_O \odot \mathbf{M}, \mathbf{X}_R \in \mathbb{R}^{P \times N}$. If the element of the mask matrix is 0, then the corresponding position in the real matrix needs to be predicted.

Definition 4 (Inference Matrix): The prediction result is denoted by $\mathbf{Y} = f(\mathbf{X}_R)$, where $f(\cdot)$ is the prediction method, $\mathbf{Y} \in \mathbb{R}^{P \times N}$.

Definition 5 (Adjacency Matrix): A matrix is constructed based on the spatial geographical location information between the sensed PoIs, used to represent the relationship between the sensed PoIs and denoted by $\mathbf{A} \in \mathbb{R}^{P \times P}$.

The frequently used notations are summarized in Table I. Our goal is to find a prediction method that minimizes the error between the inference matrix and the original matrix. The target of this paper can be expressed by the following:

$$\text{Minimize: } \sum_{t=1}^Y L(f_\phi(\mathbf{D}_W), d_t) \\ \sum_{d=1}^D L(f_\theta(\mathbf{X}_R), \mathbf{X}_O),$$

where $L(\cdot)$ is the loss function. $\sum_{t=1}^Y L(f_\phi(\mathbf{D}_W), d_t)$ aims to minimize the error associated with PoIs in tasks that are not verified by UAVs, relative to the ground truth. This is crucial for ensuring that, despite the limited number of verification tasks feasible for UAVs, the discrepancies between the predicted data and the ground truth are minimized. $\mathbf{D}_W$ represents the data exclusively provided by human participants. $f_\phi(\mathbf{D}_W)$ represents the data predicted by the model based on the data provided by human participants. And d_t represents the ground truth. The expression $\sum_{d=1}^D L(f_\theta(\mathbf{X}_R), \mathbf{X}_O)$ aims to minimize the error of the inference matrix derived from the spatio-temporal information matrix of unsensed PoIs, relative to the original matrix, within a specified time period and a certain number of sensed PoIs. This ensures that the model effectively reduces discrepancies between the inference matrix and the original one. $\mathbf{X}_R$ represents the input matrix of the model, which is the spatio-temporal matrix of unsensed data. $f_\theta(\mathbf{X}_R)$ represents the model's prediction results for the input matrix. And $\mathbf{X}_O$ represents the original spatio-temporal data matrix. Here we employ mean absolute percentage error (MAPE) and root mean square error (RMSE) to evaluate metrics:

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\%$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2},$$

where y_i and $\hat{y}_i$ are the actual and predicted values of unsensed PoIs, respectively.

IV. FEW-SHOT SAMPLES BASED TRUTH DISCOVERY METHOD

In MCS scenarios, the challenge of how to discover truth sensing data is often encountered, particularly when there are only few pieces of trusted sensing data. Few-shot samples learning methods specifically address the challenge of classification or regression with a restricted number of samples [43]. Few-shot learning is a machine learning concept where a model is trained to generalize effectively from only a small number of examples. This approach contrasts with traditional models that typically require large datasets to learn patterns effectively. In few-shot learning, the model leverages prior knowledge from similar tasks or domains to quickly adapt to new tasks with minimal data.

In this context, few-shot learning is applied to truth discovery in MCS. UAVs provide limited yet high-quality sensing data, which serve as few-shot samples. These examples are used to train a truth discovery model that enhances the accuracy of data collected by human participants. The model can adapt to new data collection tasks or environments with a small amount of UAV-sensed data, enabling efficient calibration of the data collected by human participants without the need for extensive calibration efforts.

The meta-learning is one of commonly employed few-shot samples approaches, which entails learning from diverse tasks to endow the model with prior knowledge that can be generalized to new tasks [44]. Given that obtaining limited sensing data from UAVs aligns with the few-shot learning paradigm, we propose

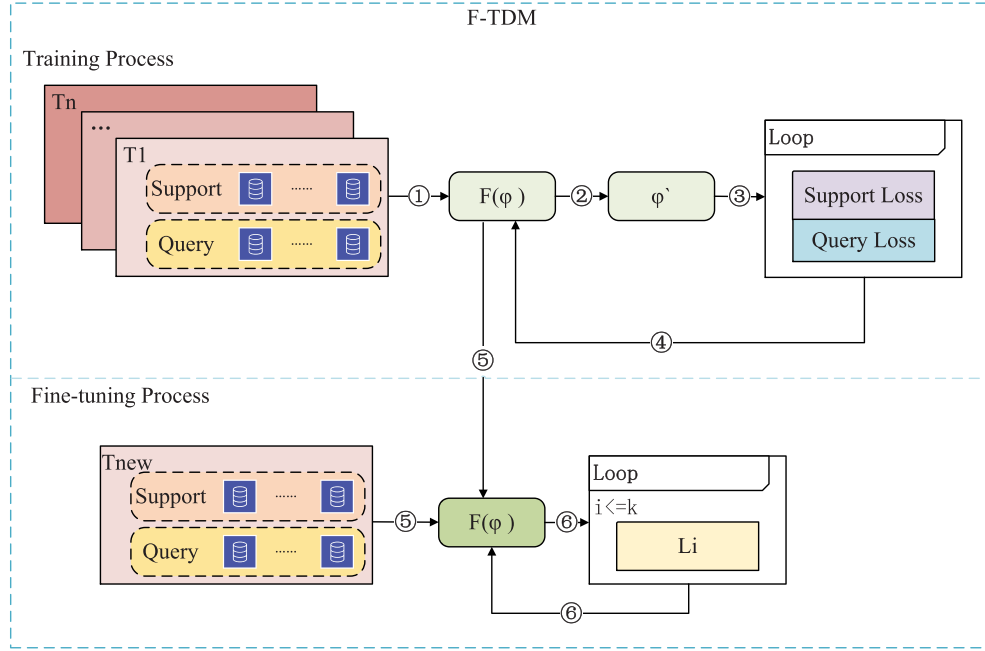


Fig. 2. Framework of the proposed F-TDM, illustrating the meta-learning process with support and query sets. The model rapidly adapts to new tasks through few-shot learning, using UAV-provided sensing data for initial calibration and fine-tuning with minimal updates. $F(\phi)$ is used to update the actual parameters of the model, while ϕ' is a copy of the parameters $F(\phi)$, used for updating the internal parameters of the model during the training process.

to utilize meta-learning methods named few-shot samples based truth discovery method (F-TDM) to address truth discovery issue of MCS. After UAVs provide few-shot samples validation data at some PoIs, F-TDM could calibrate sensing data contributed only by human participants.

Compared to few-shot learning methods like metric learning and transfer learning, F-TDM offers greater flexibility. Metric learning excels in classification tasks that involve learning similarities between samples; however, it is less suited for predicting numerical values. Transfer learning works well when source and target tasks are similar but often requires extensive fine-tuning when tasks differ. In contrast, F-TDM adapts quickly to new tasks with minimal updates and handles regression tasks without needing high data similarity, making it a more versatile option.

Fig. 2 illustrates the framework of the proposed F-TDM, which primarily involves training on similar truth discovery tasks and fine-tuning in new few-shot samples scenarios. The training process involves developing a truth discovery model on various tasks by Step 1, each comprising of a support set and a query set. The support set is employed for internal parameter updates, achieved by duplicating the current model parameters $F(\phi)$ to obtain ϕ in Step 2, followed by preliminary adjustments to ϕ' through learning from the support set samples in Step 3. Subsequently, the query set assesses the performance of the F-TDM on new samples of the same type tasks, verifying the method's effective learning and adjusting the actual model parameters. This is accomplished in Step 4. Each support or query set includes sensing data contributed by human participants and by UAVs. After multiple training rounds, Step 5 results in a model $F(\phi)$ capable of rapidly adapting to few-shot

samples scenarios. When the fine-tuning process of F-TDM occurs, the proposed method will be provided with a small number of new task-specific sensing data, each including values from human participants and corresponding UAV-provided ground truth. Through several fine-tuning updates in Step 6, the F-TDM is able to calibrate sensing data based solely on human participant contributions. By leveraging the meta-learning and few-shot learning capabilities of F-TDM, the approach enables rapid adaptation to new truth discovery tasks with limited data.

Algorithm 1 presents the training process of F-TDM. The input includes m different types of truth discovery tasks and two model hyper-parameters α and β , which are used for internal and external model updates. Each type of task is further divided into a support set and a query set, where the support set is used to update the internal parameters of the model, while the other is used to evaluate the updated model parameters. The output is the truth discovery model with the ability to quickly adapt to new tasks. The model obtains good initialization parameters through the training process.

First, all parameters of the network model θ are initialized randomly in Line 1. Then the model starts to train from Line 2 to the end. The support set $\mathcal{D}_{support}$ is extracted from the task $\mathcal{T}_i$ in Line 4. Here, $\mathbf{D}_W$ represents a total of K groups of data, where each group of data is a set of participants. $\mathbf{D}_W$ can be represented in the following matrix form:

$$\mathbf{D}_W = \begin{bmatrix} d_{1,1} & d_{1,2} & \cdots & d_{1,i} \\ \vdots & \vdots & \vdots & \vdots \\ d_{k,1} & d_{k,2} & \cdots & d_{k,i} \end{bmatrix},$$

Algorithm 1: Truth Discovery Model Training Procedure.

Input: M types of tasks $\mathcal{T}_m$, network models for regression tasks Δ , step size hyper parameters α and β
Output: the trained truth discovery model

- 1: Initialize the parameters θ of Δ randomly;
- 2: **while** not reach epochs **do**
- 3: **for** $i = 1, \dots, M$ **do**
- 4: Obtain the support dataset $\mathcal{D}_{support} = \{\mathbf{D}_W, \mathbf{D}_U\}$ from $\mathcal{T}_i$;
- 5: Use the current model to predict and obtain result $\mathbf{D}_P$;
- 6: Evaluate $L_{support_i}$ by (1);
- 7: Evaluate $\nabla_{\theta} L_{support_i}(f_{\theta})$;
- 8: Compute adapted parameters with gradient descent $\theta'_i = \theta - \alpha \nabla_{\theta} L_{support_i}(f_{\theta})$;
- 9: Obtain the query dataset $\mathcal{D}_{query} = \{\mathbf{D}_W, \mathbf{D}_U\}$ from $\mathcal{T}_i$;
- 10: Evaluate L_i with weight θ'_i ;
- 11: **end for**
- 12: Evaluate $\nabla_{\theta} L_{sum}$ and $L_{sum}(f_{\theta})$;
- 13: Update $\theta \leftarrow \theta - \beta \nabla_{\theta} L_{sum}$;
- 14: **end while**

where $d_{k,i}$ is the sensing data. $\mathbf{D}_U$ represents the validation data set provided by multiple UAVs, and its matrix form as follows:

$$\mathbf{D}_U = [d_1, d_2, \dots, d_k]^T.$$

In Line 5, we obtain the corresponding predicted values $\mathbf{D}_P = f_{\phi}(\mathbf{D}_W)$, where $f_{\phi}(\cdot)$ represents the current parameters of the model. In Line 6, the prediction error $L_{support_i}$ is calculated based on the difference between $\mathbf{D}_P$ and the ground truth values $\mathbf{D}_U$ provided by the UAV. We utilize the mean squared error loss function, which is shown as:

$$L_{support_i} = \frac{1}{K} \|\mathbf{D}_P - \mathbf{D}_U\|^2. \quad (1)$$

In Line 7, the gradient of $L_{support_i}$ with respect to the current model parameters θ is to be computed. To adapt to this task, we perform a single gradient update on the model parameters θ , as:

$$\theta'_i = \theta - \alpha \nabla_{\theta} L_{support_i}(f_{\theta}),$$

where θ'_i represents the updated parameters after the i -th task. It is observed within each individual task, the parameters are updated only once, as indicated by a single gradient update in Line 8. However, it is certainly possible to perform multiple gradient updates within each task. The purpose of F-TDM is to allow the trained parameters to be slightly updated or fine-tuned on new tasks, enabling them to quickly adapt to new tasks. The ideal situation would be to reach convergence with a single update. In order to impart the ability of model to adapt to new demands quickly, each task in the F-TDM training procedure only changes the parameters once. The performance of the updated parameters θ'_i is next tested in Line 10. After M iterations of the inner loop, we calculate the sum of errors for each iteration with respect to

the updated parameters of the current iteration in Line 12, as:

$$L_{sum} = \sum_{i=1}^M L_i(f'_{\theta_i}).$$

Finally, we update the model parameters.

After the training process, the model obtains suitable initialization parameters and has the ability to quickly adapt to new truth discovery tasks, which could achieve relatively accurate prediction results after a few fine-tunings.

V. UNSENSED POIS DATA PREDICTION METHOD

As we mentioned in Section I, the proposed method organizes the PoIs into a spatio-temporal information matrix, which is served as input for the graph neural network to grasp spatio-temporal relationships. Regarding spatial relevance, the PoIs are regarded as nodes within a graph, with the relationships between these PoIs serving as edges of the graph. Here the kriging method is employed to calculate the edge weight between each sensed PoI [45]. The kriging method is a statistical-based geological modeling method, which predicts the values at unknown locations by making optimal, unbiased estimates of known data. We utilize the Gaussian semivariogram to implement the kriging method, which can be formulated as follows:

$$\gamma(h_{ij}) = \sigma^2 \left(1 - \exp \left(-\frac{h^2}{2C^2} \right) \right),$$

where $\gamma(h_{ij})$ is the weight value between PoI i and j . σ^2 is the variance (sill), h is the distance between two PoIs, and C is a constant. After obtaining the weight values between each PoI, row normalization should be performed in order to ensure the weight contribution of each PoI is equal. Specifically, for a PoI i and j , we normalize the weight value to ensure that the sum of their weights is equal to 1. The normalization formula is shown below:

$$\gamma'(h_{ij}) = \frac{\gamma(h_{ij})}{\sum_{j=1}^P \gamma(h_{ij})},$$

where $\gamma'(h_{ij})$ is the normalized weight value, $\gamma(h_{ij})$ is the original weight value, and $\sum_{j=1}^P \gamma(h_{ij})$ is the sum of all weight values in the i -th row, while P is the total number of PoIs. Then the adjacency matrix has been constructed which provides a basis for the subsequent graph neural network model.

We combines the ideas of Graph Neural Networks with Feature-wise Linear Modulation (GNN-FiLM) [46] and residual connections [47] to implement the proposed unsensed PoI data prediction method (UPM). The main goal of GNN-FiLM is to enhance the performance of GNN when dealing with graph-structured data. FiLM is used to adjust the feature representations of nodes in each layer, allowing the model to better adapt to complex structures and patterns within the graph. By performing element-wise scaling and shifting operations on node features, GNN-FiLM allows for fine-tuning of features based on the needs of the current task. Here we select the GCN model for the FiLM transformation. Additionally, residual connections is employed to facilitate learning of complex functions by decomposing them into simpler identity mappings (residuals) and relatively smaller

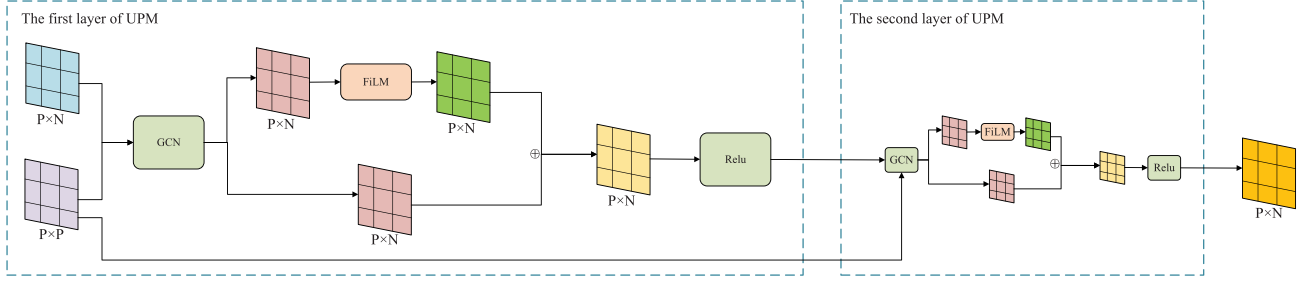


Fig. 3. Workflow of the proposed UPM model, illustrating the input of the real matrix and adjacency matrix into the GCN and GCN-FiLM layers. The model combines spatial and temporal features to predict unsensed PoIs, leveraging graph neural networks and residual connections for enhanced accuracy.

supplementary parts. Consequently, the model primarily focuses on learning these supplementary parts, which aids in approximating the target function more effectively, thereby enhancing expressive power.

Fig. 3 illustrates the workflow of the proposed UPM. The input includes a real matrix $\mathbf{X}_R \in \mathbb{R}^{P \times N}$ and a normalized adjacency matrix $\mathbf{A} \in \mathbb{R}^{P \times P}$, and the output is an inference matrix $\mathbf{Y} \in \mathbb{R}^{P \times N}$ of the same size and dimensions as the real matrix. Initially, the real matrix $\mathbf{X}_R$ and the normalized adjacency matrix $\mathbf{A}$ are input into the UPM and undergo two different types of processing. After passing through the GCN model, an initial prediction matrix is derived from these inputs. Simultaneously, the inputs are also subjected to a joint processing via the GCN and FiLM module. The GCN layer first processes the input data, then the FiLM module modulates the output of the GCN enhancing its ability to adapt to specific tasks and datasets.

Finally, the residual of the output processed by the GCN, along with the output modulated by the FiLM, are combined and fed into the activation function for nonlinear transformation. After being processed by the first layer of the UPM, a newly generated feature representation is obtained. This process is repeated, with the final hidden layer outputting features identical to the real matrix $\mathbf{X}_R \in \mathbb{R}^{P \times N}$, thereby obtaining the inference matrix $\mathbf{Y} \in \mathbb{R}^{P \times N}$. This approach ensures that the model not only learns the output modulated by GCN-FiLM but also retains the original output of the GCN. Consequently, UPM could express a richer combination of features, leading to improved performance across diverse tasks and datasets. By integrating GCN-FiLM with residual connections, UPM leverages the strengths of both architectures, resulting in a significant enhancement in predicting unsensed PoIs data.

Algorithm 2 introduces the training process of the UPM. The input consists of a series of real matrices $\mathbf{X}_R$ along with their corresponding mask matrices $\mathbf{M}$, and the adjacency matrix $\mathbf{A}$ computed using the kriging method. The output of the model is the prediction of the real matrix, represented as a series of inference matrices $\mathbf{Y}$. The training process leverages the adjacency matrix to capture the underlying spatial relationships and patterns within the data, enhancing the predictive capability of the model. In detail, we initialize the parameters randomly in Line 1. Subsequently, training epochs on the UPM are conducted from Line 2 to Line 9, where we systematically feed in a

sequence of real matrices $\mathbf{X}_R^k \in \mathbb{R}^{P \times N}$ along with the adjacency matrix $\mathbf{A}^k \in \mathbb{R}^{P \times P}$ (Line 3 and Line 4). In Real matrices $\mathbf{X}_R^k$ and adjacency matrix $\mathbf{A}^k$, k represents the data of the k -th day. This process generates the corresponding inference matrix $\mathbf{Y}^k \in \mathbb{R}^{P \times N}$. In Line 5, we evaluate the deviation by comparing the inference matrix $\mathbf{Y}^k$ to the original matrix $\mathbf{X}_O^k$, with the specific calculation detailed below:

$$L = \begin{cases} 0, & \text{if } \mathcal{Z} = 0 \\ \frac{\sum_{P,N} ((\mathbf{Y}^k[i,j] - \mathbf{X}_O^k[i,j])^2 \cdot \mu^k[i,j])}{\mathcal{Z}}, & \text{otherwise} \end{cases}, \quad (2)$$

where $\mathbf{Y}^k[i,j]$ and $\mathbf{X}_O^k[i,j]$ represent the elements of the inference and original matrix in the i -th row and j -th column, $i \in [0, P], j \in [0, N]$. $\mu^k[i,j] = 1 - \mathbf{M}^k[i,j]$, and $\mathbf{M}^k[i,j]$ represents the elements of the mask matrix in the i -th row and j -th column, $i \in [0, P], j \in [0, N]$. The index k indicates the data of the k -th day. $\mathcal{Z}$ represents the number of elements that need to be predicted, specifically the number of zeros in the matrix $\mathbf{M}^k$, that $\mathcal{Z} = \sum_{P,N} \mu^k[i,j]$. In Line 6 of the Algorithm 2, we employ the Adam gradient descent algorithm for optimizing and updating parameters of the model to minimize the error between $\mathbf{Y}^k$ and $\mathbf{X}_O^k$. Lastly, a round of training is concluded in Line 7 for the spatio-temporal information matrix that includes unsensed PoI data from D -day. Finally, we increment the training rounds to continue the training process.

After the training process, the proposed method can predict the spatio-temporal information matrix that includes unsensed PoIs for the next n days. The matrix $\mathbf{X}_R^{D+n}$ is input into the UPM, resulting in the corresponding inference matrix $\mathbf{Y}^{D+n}$. Subsequently, the original matrix $\mathbf{X}_O^{D+n}$ is merged with the inference matrix $\mathbf{Y}^{D+n}$. This process involves retaining the non-predicted data from the original matrix $\mathbf{X}_O^{D+n}$ while integrating the predicted data from the inference matrix $\mathbf{Y}^{D+n}$, resulting in the final predicted outcome $\mathbf{X}_F^{D+n}$, which could be formulated as:

$$\mathbf{X}_F^{D+n} = \mathbf{X}_O^{D+n} \odot \mathbf{M}^{D+n} + \mathbf{Y}^{D+n} \odot (1 - \mathbf{M}^{D+n}).$$

In this process, $\mathbf{M}^{D+n}$ represents the mask matrix for the spatio-temporal matrix. Here we employ the hadamard product to multiply it with the original matrix $\mathbf{X}_O^{D+n}$, preserving the data that does not require prediction. Simultaneously, we add the result of multiplying the complement of the mask matrix $\mathbf{M}^{D+n}$ with the inference matrix $\mathbf{Y}^{D+n}$ to obtain the final result matrix.

Algorithm 2: Unsensed PoIs Data Prediction Training Procedure.

Input: the real matrix set $\mathbf{X}_R$, the mask matrix set $\mathbf{M}$ and the adjacency matrix set $\mathbf{A}$
Output: the inference matrix set $\mathbf{Y}$

- 1: Randomly initialize the parameters;
- 2: **while** epoch not reach epochs **do**
- 3: **for** $k = 1, \dots, D$ **do**
- 4: Input the real matrix $\mathbf{X}_R^k$ and adjacency matrix $\mathbf{A}^k$ to obtain the inference matrix $\mathbf{Y}^k$;
- 5: Calculate L using $\mathbf{X}_O^k$ by (2);
- 6: Update the parameters of UPM to reduce L ;
- 7: **end for**
- 8: epoch = epoch + 1;
- 9: **end while**

VI. PERFORMANCE EVALUATION

In this section, we begin by introducing the real datasets utilized in the two experiments and outline the data pre-processing steps. Subsequently, we specify the evaluation metrics and comparison baselines employed in each experiment. Finally, the experimental results of the two experiments are presented individually.

A. Setup for Truth Discovery Experiment

1) *Dataset:* The study utilizes a dataset¹ consisting of 9,357 instances of hourly averaged readings collected by five metal oxide chemical sensors housed in an air quality chemical multi-sensor device, which was positioned at street level in a heavily polluted area within an Italian city. The data spans from March 2004 to February 2005, representing the most extensive publicly available records from air quality sensors deployed in real-world conditions. Hourly averaged concentrations of CO, non-methanic hydrocarbons, benzene, total nitrogen oxides (NO_x), and nitrogen dioxide (NO_2) were cross-referenced with readings from a nearby certified analyzer. The outlier data in the dataset is indicated by a constant value of -200 .

2) *Pre-Processing:* We processed the mentioned dataset as follows:

The datasets were categorized into seven distinct types of data: CO, PT08.S1, C6H6, PT08.S2, NO_2 , temperature, and NO_x . To simulate the rapid adaptability of truth discovery to new data types, we excluded the NO_x data during training and reserved it to represent a new and small-sample data type, in order to evaluate the model ability to learn new tasks in few-shot samples scenarios.

Each data type was recorded hourly. To simulate sensing data collected from both UAVs and human participants, we grouped continuous data into sets of five, consisting of data from four human participants and one UAV. The outlier values were set to -200 . And several sets might have one or two outlier sensing data that contributed by human participants.

¹<https://www.kaggle.com/datasets/aayushkandpal/air-quality-time-series-data-uci>

TABLE II
LIST OF TRUTH DISCOVERY EXPERIMENTAL PARAMETERS

Parameters	Value
Number of inner loop update	1
Hidden size	1024
Inner loop learning rate	0.0005
Outer loop learning rate	0.0005
Number of training epochs	5000

During the experiment, a small amount of new type data was introduced to simulate new tasks performed under conditions of limited ground truth availability. For the new NO_x data type, we provided the model with only a few data (i.e., 6, 7, 8, 9, 10) for learning. We then evaluated the performance of the proposed model using various numbers of PoIs, which only included data contributed by human participants. The number of test samples started at up to ten times the number of learning PoIs.

3) *Experimental Environment:* The experiment was conducted on an Ubuntu 18.04.3 x64 server, equipped with an Intel(R) Xeon(R) Gold 5122 CPU (4 cores @ 3.60 GHz), 62 GB of memory, and 4 Nvidia GeForce RTX 2080Ti graphics cards. The proposed methods are implemented using Python 3.7 and PyTorch 1.2.0.

To enable the model to rapidly adapt to new tasks, the number of inner loop updates was set to 1. The specific experimental parameters include a hidden size of 1024, with both the inner and outer loop learning rates were set to 0.0005, and the number of training epochs was set to 5000. These parameters are detailed in Table II.

4) *Evaluation Metrics:* To evaluate the performance of our method, we conducted an accuracy assessment of the truth discovery method in a small sample scenario. For this purpose, we employed RMSE as the evaluation metric: $RMSE = \sqrt{\sum_{i=1}^n (y_i - \hat{y}_i)^2 / n}$, where y_i is the actual value of PoI i and $\hat{y}_i$ is the predicted value.

5) *Control Experiment:* To compare the performance of our method, we extracted data from a new type dataset to simulate a few-shot samples scenario. Subsequently, we compared our proposed method with four baselines:

- 1) Random parameters: this baseline represents a method without prior learning, employing random parameters with the same model and learning rate as F-TDM.
- 2) Incentive-based Truth Discovery (ITD) [12]: the method is employed in handling truth predictions for data exclusively submitted by human participants. In the weight estimation phase, values provided by the UAVs serve as the ground truth for updating the weights of human participants. During the truth inference phase, the estimated weights of the human participants are used to calculate a weighted aggregation calculation on the new sensing data.
- 3) Privacy-preserving Truth Discovery with Outlier Detection (TDOD) [48]: this scheme incorporates outlier detection alongside weight and truth update phases to predict the truth for data solely submitted by human participants. During the weight update phase, human participant weights are adjusted based on ground truth information.

Subsequently, in the truth update phase, outlier detection is initially conducted. Identified outliers are eliminated before the updated weights are employed to determine the truth for new sensing data. This comprehensive approach ensures that the truth discovery process is robust against outliers, thereby enhancing the accuracy and reliability of the predicted truths derived from sensing data contributed by human participants.

- 4) Mean algorithm: traditional truth discovery methods that lack knowledge of the trustworthiness of human participants in a small-sample scenario. Therefore, here all the human participants are assumed the same level of trust.

These baselines serve as benchmarks for evaluating the effectiveness of our proposed method.

B. Setup for Unsensed PoI Prediction Experiment

1) *Dataset*: The dataset utilized in the unsensed PoIs data prediction experiment comprises air quality data from air quality monitoring stations in Beijing and Shanghai, spanning from February 8, 2013, to February 8, 2014. Each dataset for the two cities contains information about city base stations and air quality. The base station information includes the station's ID and the latitude and longitude coordinates of the base station. The air quality information for each base station encompasses data of various types over a series of time periods, including PM2.5, PM10, NO₂, humidity, and other types of data.

2) *Pre-Processing*: In the experiments detailed in this paper, we selected the data from Shanghai and processed it according to the following steps:

We structured data of the same type from various time segments at different base stations on a given day into a spatio-temporal information matrix. Each row represented data from different time segments at a specific base station, while each column represented data from different base stations at a particular time segments.

In our experiment on predicting unsensed PoIs data, we evaluated three distinct datasets: humidity, PM2.5, and PM10. These datasets contained available data spanning 123, 120, and 121 days, respectively. We retained the data from the first 120 days and split it into training and test sets at an 8:2 ratio. Each test set comprised 24 days, and the average results over these 24 days were considered the final evaluation metric. Data of each day consisted of readings from 10 base stations across 20 time segments. To evaluate the predictive capability of the model under varying degrees of available information, we conducted experiments for each test set where the known PoIs represented 10%, 20%, 30%, 40%, 50%, and 60% of the total. This comprehensive approach allowed us to evaluate the performance of the model across a range of scenarios with differing levels of available data.

To thoroughly evaluate the performance of the UPM, we conducted a detailed ablation experiments. The objective was to compare the performance disparities between GCN-FiLM and UPM across different test sets. These ablation experiments followed the same data processing approach as the unsensed PoIs data prediction experiment and was carried out on the humidity, PM2.5, and PM10 test sets.

TABLE III
LIST OF UNSENSED POI PREDICTION EXPERIMENTAL PARAMETERS

Parameters	Value
Learning rate	0.0001
Hidden size	256
Dropout rate	0.5
Number of epochs	200
Number of PoIs	10
Time segments	20

3) *Experimental Environment*: The experimental environment for these experiments remained consistent with that utilized for the truth discovery experiments previously mentioned.

The experiment was conducted to use a two-layer of UPM. To ensure fairness in the comparative experiments, all methods used the same learning rate of 0.001, a dropout rate of 0.5, a hidden size of 256, and 200 training epochs. The number of PoIs was set to 10, and the time segment to 20. The specific parameters are detailed in Table III.

4) *Evaluation Metrics*: In addition to RMSE, we also utilized MAPE to assess the experimental results: $MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\%$, where y_i and $\hat{y}_i$ were the actual and predicted values of unsensed PoIs, respectively.

5) *Control Experiment*: To assess the effectiveness of our method, we conducted experiments using diverse datasets, encompassing humidity, PM2.5, and PM10. We compared our proposed approach with three baseline methods:

- 1) Graph Convolutional Networks (GCN): GCN operates on graph structures, employing convolution operations to capture spatial correlations among PoIs. We utilized GCN to model these correlations and predict unsensed PoIs. The adjacency matrix generation followed the same methodology as our proposed method (UPM), ensuring equitable comparison. Furthermore, training and testing procedures were consistent with those of UPM.
- 2) Graph Attention Networks (GAT): GAT excels in discerning node importance within graph-structured data, thereby capturing intricate patterns. Analogous to GCN, we employed GAT to forecast unsensed PoIs, with adjacency matrix generation and training/testing procedures mirroring those of UPM.
- 3) Bidirectional Gated Recurrent Unit (Bi-GRU): Bi-GRU, a variant of recurrent neural networks, adeptly handles long-term dependencies, making it suitable for time series prediction tasks. We utilized a bidirectional Bi-GRU to predict unsensed PoIs, conducting training and testing on identical datasets as UPM.

Through comparative analysis across these baselines and datasets, we assessed the effectiveness of the proposed method in addressing the challenge of predicting unsensed PoIs data within MCS scenarios.

C. Experimental Results of Truth Discovery

In order to verify the rapid learning capability of F-TDM in new few-shot samples scenarios, experiments were conducted on the NO_x dataset with F-TDM and the baselines. To highlight the characteristics of small sample scenarios, for new truth

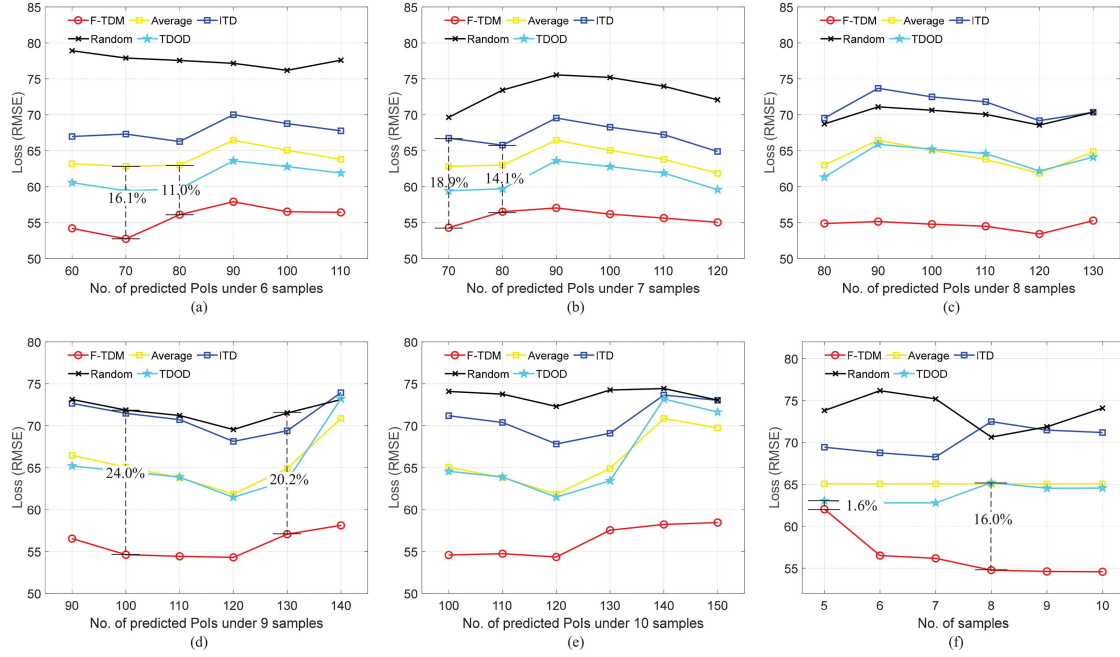


Fig. 4. Simulation results, where (a)–(e) are the comparison of prediction errors under 6–10 samples, respectively. (f) shows the comparison of prediction errors with the various number of samples when there are 100 PoIs to be predicted.

discovery task, we provided the model with 6, 7, 8, 9, 10 PoIs respectively for learning, and then tested it with different numbers of PoIs to perform the quick adaptability of the model. The number of test PoIs started at a minimum of ten times the number of learning PoIs, subsequently increased incrementally by 10. Fig. 4 showed the results of the experiment. In terms of prediction error, F-TDM regularly outperformed all baselines. After training, the running time of F-TDM is influenced by the volume of sensing data contributed by UAVs. Consequently, the time complexity of the trained F-TDM can be expressed as $\mathcal{O}(n)$ multiplied by $\mathcal{O}(\text{Model})$, where $\mathcal{O}(\text{Model})$ represents the time complexity of the specific model used in the F-TDM method, and n denotes the amount of sensing data contributed by UAVs. In this paper, the model's time complexity is $\mathcal{O}(\text{hidden} * \text{Max}(\mathcal{I}, \text{hidden}))$, where $\mathcal{I}$ is the number of human participants at the same PoI, and hidden denotes the size of the hidden layer. Therefore, the overall time complexity of the trained F-TDM is $\mathcal{O}(n * \text{hidden} * \text{Max}(\mathcal{I}, \text{hidden}))$.

For instance, it was observed in Fig. 4(a) that after learning 6 sensed PoIs, F-TDM performed better than other baselines when the number of PoIs ranged from 60 to 110 in quantity. Specifically, when the number of PoIs were 70 and 80 respectively, the accuracy of F-TDM was 16.1% and 11.0% higher than that of the average algorithm. In Fig. 4(b), F-TDM outperformed TDOD by 18.9% and 14.4% when the number of PoIs were 70 and 80, respectively. In Fig. 4(d), F-TDM consistently outperformed the other baselines in terms of prediction accuracy for all PoIs. When the number of PoIs were 100 and 130, respectively, F-TDM exceeded the Random algorithm by 24.0% and 20.2% in terms of accuracy.

Finally, we tested the prediction results of methods under different sensed PoIs conditions for the same PoIs sensed all

by human participants. The experiment was set 100 identical PoIs, and all the methods learned sensed PoIs from 5 to 10. As shown in Fig. 4(f), in all instances, F-TDM demonstrated superior prediction accuracy compared with the other baselines. Upon learning 5 and 8 sensed PoIs respectively, the accuracy of F-TDM surpassed that of TDOD by 1.6% and 16.0%.

Additionally, experiments were conducted to evaluate the additional costs and data quality in scenarios involving UAVs participation. The experiments were based on the DJI Mavic 3 Pro, with relevant parameters sourced from the DJI official website.² To ensure accuracy in assessing additional parameters, the experiments were carried out under the most stringent conditions, specifically set as follows:

- 1) The maximum transmission power of 33 dBm was used for UAV communication, as selected from regulatory standards by the Federal Communications Commission (FCC), Conformité Européenne (CE), State Radio Regulatory Commission (SRRC), and Ministry of Internal Affairs and Communications (MIC). At this power level, the UAV has a maximum flight time of 43 minutes and a maximum hovering time of 37 minutes.
- 2) The minimum effective communication distance of 8 kilometers was selected, based on FCC (15 kilometers), CE (8 kilometers), SRRC (8 kilometers), and MIC (8 kilometers).
- 3) The longest single data collection time across various sensors was used to determine the UAV's hovering duration.
- 4) The server power was configured to match the experimental server setup, utilizing the maximum power output of a single GPU.

²<https://www.dji.com/hk-en/mavic-3-pro/specs>

The Fermat-Torricelli points are essential for minimizing the average single-flight distance between each PoI. However, the distance from a Fermat-Torricelli point to any PoI must fall within the UAV's effective communication range. In many cases, a single Fermat-Torricelli point cannot meet this requirement. Therefore, multiple Fermat-Torricelli points are identified to ensure that the distance to any PoI remains within the UAV's effective communication range. This can be achieved using the Weiszfeld algorithm [49]. Once the latitude and longitude coordinates of the Fermat-Torricelli points are established, the Haversine formula is employed to calculate the distances between these points and all PoIs. The average of these distances is then used to estimate the UAV's single-flight distance. The detailed process is outlined as follows:

The Weiszfeld algorithm is an iterative method to find the Fermat-Torricelli point. The specific steps are as follows:

- 1) Initialization: First, choose the centroid of all PoIs as the initial guess. The coordinates of the centroid are:

$$X_0 = \frac{\sum_{i \in \mathcal{P}} X_i}{|\mathcal{P}|}, \quad Y_0 = \frac{\sum_{i \in \mathcal{P}} Y_i}{|\mathcal{P}|},$$

where $\mathcal{P}$ is the set of PoIs. The longitude and latitude of the i^{th} PoI is denoted by X_i and Y_i , respectively.

- 2) Iterative Calculation: In each iteration, update the coordinates of the current Fermat-Torricelli point. For the k -th iteration, the new Fermat-Torricelli point P_{k+1} is calculated as:

$$P_{k+1} = \begin{cases} P_k, & \text{if } \sum_{i \in \mathcal{P}} \frac{P_i}{\|P_k - P_i\|} = 0 \\ \frac{\sum_{i \in \mathcal{P}} \frac{P_i}{\|P_k - P_i\|}}{\sum_{i \in \mathcal{P}} \frac{1}{\|P_k - P_i\|}}, & \text{otherwise} \end{cases}$$

- 3) Convergence Condition: Repeat the iteration until the distance between P_{k+1} and P_k is less than a predefined threshold.
- 4) Coverage Range Check: After calculating the Fermat-Torricelli point, check if the distance from this point to all PoIs is less than the effective communication distance. This process utilizes the Haversine formula to calculate the spherical distance between two latitude and longitude coordinates, which can be expressed as $R \cdot c$, where R is the Earth's radius and c is the angular distance between the two points, that:

$$c = 2 \cdot \text{atan2} \left(\sqrt{a}, \sqrt{1-a} \right),$$

$$a = \sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2 \left(\frac{\Delta\lambda}{2} \right),$$

where $\Delta\phi$ is the latitude difference between the Fermat-Torricelli point and the PoI, with ϕ_1 and ϕ_2 representing the latitudes. Similarly, $\Delta\lambda$ is the difference in longitude between the Fermat-Torricelli point and the PoI, with λ_1 and λ_2 representing the longitudes. The Earth's mean radius R is taken as 6371km, and atan2 refers to the arctangent function, ensuring the result falls within the interval $[-\pi, \pi]$.

The experimental data was obtained from the actual PoIs used in the Unsensed PoI Prediction experiment described in

Section VI-B. Using the aforementioned formula, the average single-flight distance of the UAV from the Fermat-Torricelli point was calculated to be 5.45 kilometers. Following this, the UAV's power consumption during the round-trip flight and data collection was computed. This was then combined with the server's power consumption over the same period to determine the total energy consumption. The electricity cost was calculated based on the weighted average of peak and off-peak rates in Shanghai. Assuming the server controls the UAV flight and UAV depreciation costs are excluded, the cost for a single UAV deployment was estimated at 0.06 RMB per UAV. The relationship between the additional operational cost of the UAV and the improvement in data quality is presented in Table VI.

D. Experimental Results of Unsensed PoI Prediction

To verify the effectiveness of UPM, the unsensed PoIs data prediction experiments were conducted on different datasets: humidity, PM2.5, and PM10. After training, UPM directly predicts data, with its runtime determined by the GCN and FiLM components. The time complexity of GCN is $\mathcal{O}(P * N * \text{hidden}) + \mathcal{O}(P^2 * \text{hidden})$, while the time complexity of FiLM is $\mathcal{O}(P^2 * \text{hidden}) + \mathcal{O}(P * \text{hidden}^2)$, where P is the number of PoIs, N is the number of time segments, and hidden denotes the size of the hidden layer. Consequently, the overall time complexity of UPM is $\mathcal{O}(\text{hidden} * P * \text{Max}(P, N, \text{hidden}))$.

Fig. 5 showed that UPM generally outperformed other baselines according to RMSE and MAPE metrics on different datasets. For instance, it was observed in Fig. 5(a) that UPM outperformed other baselines in all cases on the humidity dataset under the RMSE evaluation metric. When the ratios of sensed PoIs were 30% and 60%, respectively, UPM improved the prediction accuracy by 62.2% and 33.1% compared with GCN. In Fig. 5(b), on the PM10 dataset, when the ratios of sensed PoIs were 60%, UPM improved the prediction accuracy by 16.7% compared with Bi-GRU. Fig. 5(d)–(f) represented the performance under the MAPE evaluation metric for different datasets. For example, the GAT did not perform as well as UPM in all cases. Although the performance of Bi-GRU was close to UPM on the humidity dataset, it fell short in comparison with UPM on the PM10 dataset, and there were significant fluctuations compared with UPM on the PM2.5 dataset. Both GCN and UPM presented a good downward trend, but in most cases, the performance of GCN was inferior to UPM. As seen in Fig. 5(f), when the ratios of sensed PoIs were at 30% and 60%, UPM showed an increase in prediction accuracy by 42.4% and 21.2%, respectively, in comparison with GAT. Simultaneously, from Fig. 5 we observed that as the ratios of sensed PoIs increased, the prediction error of UPM showed a downward trend in all cases. It not only demonstrated UPM learning ability, but also reflected the stability of the proposed method.

Finally, we conducted detailed ablation experiments on humidity, PM2.5 and PM10 datasets using RMSE and MAPE evaluation metrics, testing the effectiveness of the modifications made to GCN-FiLM. The ratios of sensed PoIs for each experiment were set from 10% to 90%. The experimental results were shown in Tables IV and V. By comparing the performance

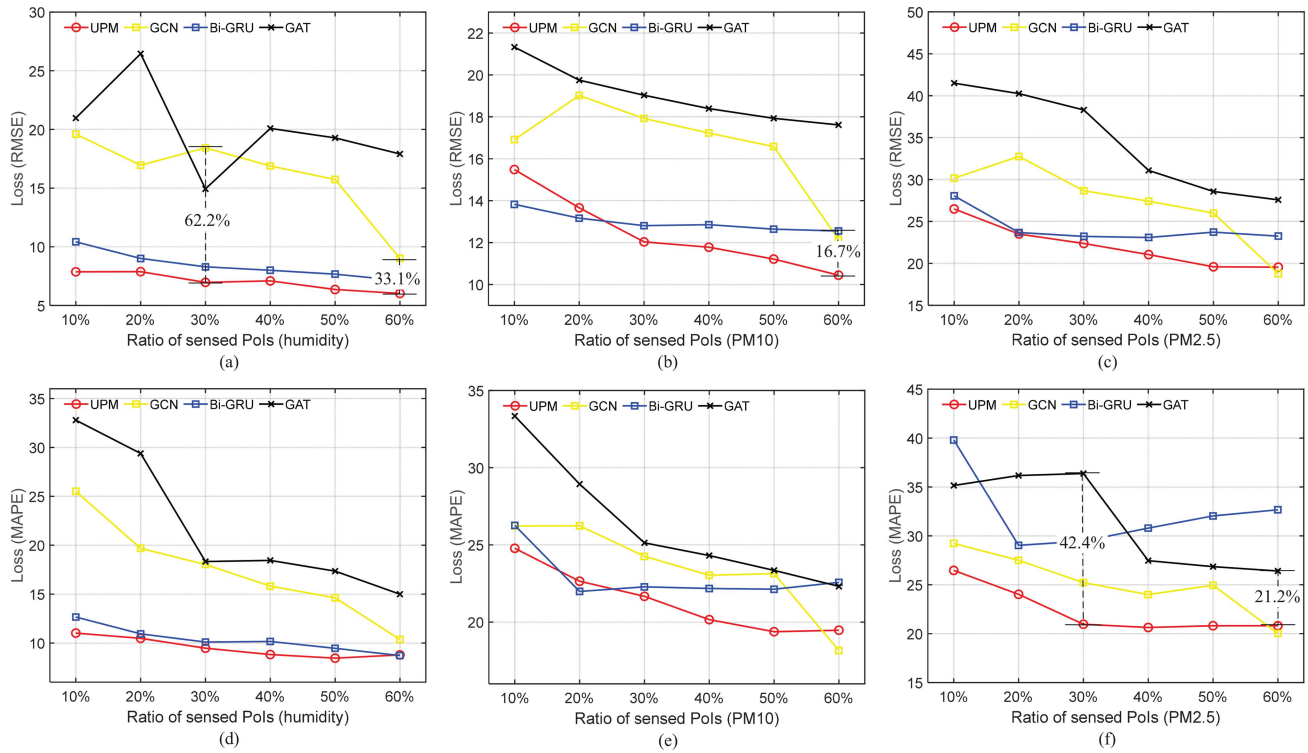


Fig. 5. Simulation results of the proposed UPM and the compared methods, where (a)–(c) show RMSE metric results and (d)–(f) present MAPE results.

TABLE IV
ABLATION EXPERIMENTS OF RMSE PERFORMANCE COMPARISON OF DIFFERENT DATASETS AT VARYING RATIOS

Data	humidity		PM10		PM2.5	
Method	UPM	GCN-FiLM	UPM	GCN-FiLM	UPM	GCN-FiLM
10%	7.8651	23.8236	15.4855	24.6004	26.4958	54.2368
20%	7.8780	23.4260	13.6565	25.2017	23.5026	55.3712
30%	6.9647	25.0202	12.0331	28.2131	22.3795	55.0610
40%	7.0916	23.1773	11.7777	27.7974	21.0442	54.8322
50%	6.3567	23.0648	11.2106	28.6963	19.5947	56.9908
60%	6.0240	23.1409	10.4475	27.9165	19.5438	57.7942
70%	5.8407	23.6780	10.0483	28.3629	18.6468	55.6772
80%	5.8491	25.5479	9.8117	30.0991	18.3642	56.3759
90%	6.1076	30.4069	9.2460	32.9976	18.7737	58.4102

TABLE V
ABLATION EXPERIMENTS OF MAPE PERFORMANCE COMPARISON OF DIFFERENT DATASETS AT VARYING RATIOS

Data	humidity		PM10		PM2.5	
Method	UPM	GCN-FiLM	UPM	GCN-FiLM	UPM	GCN-FiLM
10%	11.0191	21.1595	24.7792	36.8446	26.4658	46.4049
20%	10.4832	20.3198	22.6435	38.0760	24.0251	46.9600
30%	9.4670	23.2431	21.6723	43.8610	20.9576	45.1735
40%	8.8255	20.2980	20.1584	42.5365	20.6356	45.2321
50%	8.4593	20.2631	19.3782	43.7107	20.8161	47.2940
60%	8.7869	20.7460	19.4753	41.7367	20.8105	47.4707
70%	8.5295	21.6952	19.9034	42.4987	21.9821	46.2448
80%	8.2957	22.3468	19.9573	44.4834	23.3401	46.8611
90%	8.3649	27.3030	18.5976	45.3780	28.8718	56.1506

TABLE VI
LIST OF TRUTH DISCOVERY EXPERIMENTAL PARAMETERS

Operational Cost (RMB)	Data Quality Improvement (%)
5 * 0.06	2.26
6 * 0.06	6.46
7 * 0.06	6.72
8 * 0.06	7.78
9 * 0.06	7.91
10 * 0.06	7.93

of the two methods on the three test sets, we found that UPM outperformed the benchmark model GCN-FiLM in all cases. For instance, it was observed in Table IV, on the humidity dataset under the RMSE evaluation metric, the prediction accuracy of UPM improved by at least 66.4% and at most 79.9% compared with GCN-FiLM, in the cases where the ratios of sensed PoIs were 20% and 90%, respectively. On the humidity, PM10, and PM2.5 datasets, compared with GCN-FiLM, UPM prediction accuracy had improved by an average of 72.6%, 58.4%, and 62.6%, respectively. The experimental results under the MAPE evaluation metric for different datasets were shown in Table V. It was observed that on the PM2.5 dataset, the UPM prediction accuracy improved by at least 43.0% compared with GCN-FiLM when the ratios of sensed PoIs were at 10%, and improved by a maximum of 56.2% when the ratios of sensed PoIs were at 60%. On the three different datasets, the prediction accuracy of UPM improved by an average of 57.9%, 50.3%, and 51.5% respectively, compared with GCN-FiLM.

VII. CONCLUSION AND FUTURE WORK

In this paper, we introduced a UAV-assisted high-quality sensing data coverage method to deal with uncontrollable data quality and inevitable unsensed PoIs problems. Specifically, we proposed a novel few-shot samples based truth discovery method called F-TDM to mitigate issues arising from unknown credible human participants in scenarios with limited UAV collected sensing data. F-TDM underwent initial training across multiple tasks to adapt swiftly to new tasks, requiring only minimal fine-tuning to achieve reliable predictive performance. Our experiments across various small-sample tasks demonstrated that F-TDM consistently outperformed baseline methods. To address the challenge of unsensed PoIs, the UPM utilized the kriging method to obtain spatial correlations among PoIs and combined this information with historical sensing data to predict unsensed PoIs data values. Our experiments across diverse datasets confirmed the effectiveness of UPM compared to other baseline methods, thereby validating the proposed modifications.

In the future, we plan to establish a library containing a variety of meta-models tailored to different MCS application scenarios, such as regression meta-models for environmental monitoring and classification meta-models for traffic pattern recognition. Based on the specific MCS task, the platform will automatically select the appropriate model from the meta-model library for training, effectively addressing few-shot sample challenges.

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